BIOT 6900 Module 6

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Introduction

The goal of this module was to examine how well Alzheimer's Disease (AD) polygenic risk scores (PRSs) that were gathered using European data function across different groups of people from different backgrounds. This was done by examining genotypes from the 1000 Genomes Project across five ancestry groups, applying the score from PGS000334[3]. Most genomic data has historically been collected using European subjects[1], meaning there is potential bias when examining genetic risk. One important caveat is that the phenotypes are simulated, since the 1000 Genomes Project does not include disease labels. Therefore, AUC and calibration results are there for example only.

Data and Methods

To begin, the proof of real data:

```
source      : REAL
people      : 2336
score size  : 22 variants
by ancestry : {'AFR': np.int64(500), 'EAS': np.int64(500), 'EUR':
      np.int64(500), 'SAS': np.int64(489), 'AMR': np.int64(347)}
   genotypes.csv: sha256 9cf41a1f
   variants.csv: sha256 f7a87d3f
   score.csv: sha256 7f80e8ff
   meta.csv: sha256 6cc55b6e
```

The data was then harmonized. This was done by keeping the variants that were labeled ALT, flipping the alleles labeled REF, and dropping the variants that were labeled neither of the prior. The alleles were also checked for those that did not appear in the panel. As a result, there were 9 ALT-matching variants, 13 REF-matching variants (which were flipped), no variants dropped for mismatch, and no missing variants. A majority of the variants needed to be flipped, highlighting the importance of the harmonization, given the evidence would have been in the wrong direction for those alleles.

After the harmonization, the PRS was calculated as the genotype matrix G multiplied by the weight vector w, and then standardized. This data was evaluated by per-ancestry histograms, top-10% cutoff analysis, AUC, a calibration model fit on EUR, and a cross-validated AUC for age vs age+PRS.

Results

First, the distribution shift among the ancestry groups was found and shown below in Table 1:

Table 1

ancestry	n	mean_prs	pct_above_top10_cutoff
EUR	500	0.036	7.8
AFR	500	0.744	28.0
EAS	500	-0.472	2.2
SAS	489	-0.245	4.3
AMR	347	-0.099	6.3

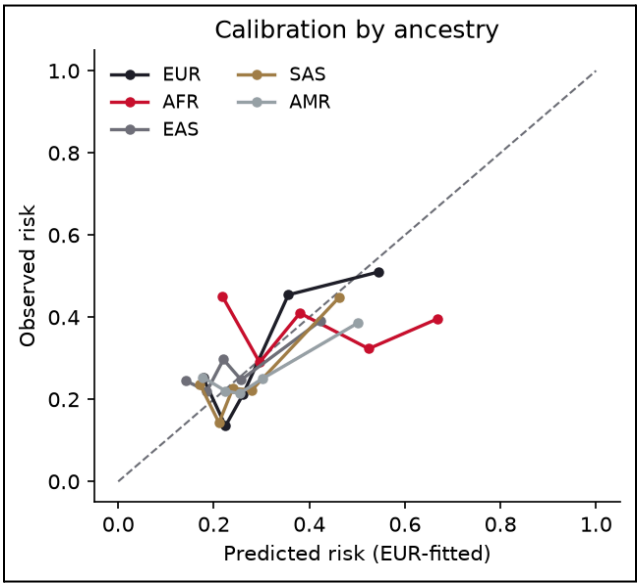
This shows a mean PRS that is highest for AFR, at 0.744, and lowest for EAS, at -0.472. The 10% cutoff is most accurate for EUR, while it is the least accurate for AFR, which has 28% above the cutoff. Then, the discrimination by ancestry was found by the AUC, with overall AUC being 0.599. The by-ancestral group data is shown below in Table 2:

Table 2

	EUR	**SAS**	**EAS**	**AMR**	**AFR**
AUC	0.672	0.611	0.575	0.555	0.493

This shows a high AUC of 0.672 for EUR, decreasing for SAS, EAS, and AMR, with the lowest for AFR at 0.493. Then, the calibration was done for EUR compared to the other ancestry regions, shown in Figure 1:

Figure 1



Finally, age only gives an AUC of 0.562 and age+PRS gives an AUC of 0.618. The age+PRS is higher by 0.056.

Discussion

LD differs significantly between different ancestry regions[2]. Because the variants and weights were calculated for mostly European subjects, the accuracy is reduced for other ancestries. The drop in accuracy shown in the AUC aligns with this idea. The allele frequencies also differ between the groups for reasons other than Alzheimer's, which shifts the distributions shown in Figure A1.

The 10% cutoff is not fair for all groups of people. For those in the AFR group, 28% of people were above the cutoff, meaning that many more people would be flagged incorrectly for being at risk for AD. On the other side of the spectrum, the test would not be sensitive for people in the EAS group, given that only 2.2% of people were above the 10% threshold. Both are unwanted outcomes, given that people not at risk in the AFR group would be caused unnecessary stress and extra testing and people at risk in the EAS group would be more likely to be missed by the test.

The AUC tests how likely it is for the PRS to be higher for someone with AD vs without AD. Given this, the PRS is the best indicator for EUR, then decreases for SAS, EAS, AMR, and AFR, in that order. EUR has a modestly decent AUC of 0.672, but the rest are between 0.611 and 0.493. With 0.5 being an even chance of PRS being higher for someone with AD vs without, these numbers are fairly weak and show the PRS may not be a good predictor of AD risk for those groups. The calibration, shown in Figure 1, shows a somewhat different story. The best overall trend, always increasing observed risk with calculated risk, was in the EUR group, as is expected. However, SAS, EAS, and AMR stay fairly close to the observed risk = calculated risk line, sometimes even closer than EUR. In contrast, the AFR line matches neither the overall trend nor the values for the most part, showing that the PRS found this way is not a good predictor for the AFR group's risk of AD. Therefore, SAS, EAS, and AMR scores disagree somewhat between discrimination and calibration.

This data could help for patients that specifically belong to the EUR group, given the results were the most promising for this group. However, given that even in this group the results were moderately successful at best, it should be used in conjunction with other tests to ensure accuracy. If it were used for people from other ancestry groups, it would need to be recalibrated and reevaluated to see if this data is useful. The increase in AUC from age to age + PRS was only 0.056, which is not large. Real phenotype data, including data from other ancestries, would need to be included in order for this data to provide real value.

Since the proxy for ε4 is not closely linked with it in the AFR group, the score is not effective for this group. This doesn't mean that ε4 is unimportant for this group, just that given the nature of

the data, it is scored incorrectly. In fact, it has been shown to be a risk factor across ancestral groups[4].

Conclusion

Given the data, the score works moderately well for EUR, whereas it does not work very well for the other ancestry groups. A single threshold does not work for every group, and is therefore not a fair cutoff to use universally among the groups or in general. Finally, to use this data in a clinical setting, more data including other ancestral groups and phenotypes needs to be collected.

Sources

[1] Martin AR, Kanai M, Kamatani Y, Okada Y, Neale BM, Daly MJ. Clinical use of current polygenic risk scores may exacerbate health disparities. Nat Genet. 2019;51(4):584-591. doi:10.1038/s41588-019-0379-x

[2] Duncan L, Shen H, Gelaye B, et al. Analysis of polygenic risk score usage and performance in diverse human populations. Nat Commun. 2019;10(1):3328. doi:10.1038/s41467-019-11112-0

[3] Zhang Q, Sidorenko J, Couvy-Duchesne B, et al. Risk prediction of late-onset Alzheimer's disease implies an oligogenic architecture. Nat Commun. 2020;11(1):4799. doi:10.1038/s41467-020-18534-1

[4] Belloy ME, Andrews SJ, Le Guen Y, et al. APOE genotype and Alzheimer disease risk across age, sex, and population ancestry. JAMA Neurol. 2023;80(12):1284-1294. doi:10.1001/jamaneurol.2023.3599

Appendix

Figure A1

